

DEPARTMENT OF APPLIED MATHEMATICS AND STATISTICS

YEAR 4 | DATA VISUALIZATION COURSE

FINAL REPORT

Course: Data Visualization

Dashboard Design and Using Microsoft Power BI

Project Title: Superstore Regional Sales Performance Analysis

Group Name / Number: Group 3

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Semester / Academic Year: 2025-2026

Submission Date: June 16, 2026

Abstract

This report presents an interactive regional sales performance dashboard built using Microsoft Power BI, utilizing the Superstore retail dataset (2020-2023). The primary objective is to transform static transactional data into dynamic business intelligence, allowing management to track Year-over-Year (YoY) growth across Total Revenue, Profit, and Quantity. By implementing a Star Schema and utilizing DAX time-intelligence functions, the dashboard provides a clear comparison of geographic regions and customer segments. Key findings reveal that the Central region is the dominant leader in overall volume and profit, while the West region demonstrates the most consistent year-over-year growth. Conversely, the South region exhibits vulnerabilities due to an over-reliance on the Corporate customer segment. The resulting visualization equips stakeholders with the actionable insights necessary to optimize regional strategies, adjust discount policies, and prepare for Q4 seasonal demand surges.

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1. Introduction

1.1 Background and Motivation

In the modern retail environment, data-driven decision-making is critical for maintaining profitability and optimizing operations. This project focuses on analyzing the sales performance of a retail enterprise using the Superstore dataset. By transitioning from static, traditional reporting to an interactive Business Intelligence (BI) environment, the project aims to transform raw transactional data into actionable insights. The resulting analysis provides a clear view of operational efficiency, regional performance, and demographic trends, equipping management with the tools necessary to make informed strategic decisions.

1.2 Problem Statement

Retail managers currently lack a clear and interactive method to monitor regional sales performance, track growth trends, and compare customer segments across different time periods. Specifically, the organization faces the following analytical bottlenecks: difficulty comparing sales performance across multiple fiscal years (2020-2023), unclear customer segment visibility without proper visual representation, and inefficient traditional reporting that fails to provide dynamic filtering.

1.3 Objectives

- Process and clean historical retail data to ensure analytical accuracy.
- Visualize Year-over-Year (YoY) growth trends for key performance indicators, including Total Revenue, Total Profit, and Total Quantity.
- Illustrate the financial contribution of various customer segments segmented by geographic region.
- Provide stakeholders with an interactive, user-friendly interface to support immediate, data-backed operational decisions.

1.4 Scope and Limitations

This project analyzes the Superstore sample dataset covering retail sales transactions in the United States over a four-year period (2020-2023). The analysis encompasses four regions (West, East, Central, South) and customer segments. Limitations include the lack of deeper geographic breakdown (state/city level), the absence of cost structure data beyond profit, and the restriction of analysis variables to only those available within the dataset.

2. Data Description

2.1 Dataset Overview

Describe the dataset: source, format, size, and time period covered.

The secondary data used for this project is the "Superstore Regional Sales" dataset, sourced from Kaggle. It serves as a representative sample of local business retail operations, containing historical transactional data spanning four years (2020-2023). The dataset encompasses a variety of business dimensions, including geographic regions, customer classifications, and core financial metrics, making it highly suitable for comprehensive retail analytics.

2.2 Data Dictionary

Variable Name	Data Type	Unit	Description
Order ID	Whole Number	-	Unique identifier for each transaction
Order Date	Date	-	The date the customer placed the order
Customer Name	Text	-	Full name of the customer
Customer Segment	Text	-	Classification of the buyer (e.g., Corporate, Consumer, Home Office)
Region	Text	-	Geographic location of the sale (Central, East, South, West)
Product Category	Text	-	Broad grouping of the item sold
Quantity	Whole Number	Units	Total number of items purchased in the order
Unit Price	Whole Number	USD	The price charged for a single unit of the product
Revenue	Decimal Number	USD	Total sales amount generated from the transaction
Profit	Decimal Number	USD	Net money gained from the sale after costs
Product Margin	Whole Number	%	The profit margin percentage of the item
Month_Name / Year	Text / Whole No.	-	Time attributes extracted for dashboard slicers and charts

2.3 Data Preprocessing

Before building the dashboard, the data was systematically cleaned and prepared in Power Query Editor to ensure accuracy and consistency.

- **1. Data Cleaning & Formatting:** Identified and handled missing or null values. Scanned and removed duplicate transaction rows to ensure accurate totals. Standardized Data Types: Quantity (Whole Number), Unit Price & Product Margin (Decimal/Percentage), Order Date (Date format).
- **2. Fact Table Enhancements (Calculated Columns):** Created new columns directly in the Fact Table for accurate row-level evaluation:
 - Revenue = 'Order Table'[Unit Price] * 'Order Table'[Quantity]
 - Profit = 'Order Table'[Revenue] * 'Order Table'[Product Margin]
 - Extracted time attributes: Year, Month_Number, and Month_Name.
- **3. Data Modeling:** Created a dedicated Date Table using = CALENDARAUTO() to support dynamic time-intelligence functions on the main overview dashboard.

3. Dashboard Design

3.1 Design Principles and Rationale

The dashboard was designed following the Star Schema principle. The layout was structured as a single-page interactive report with clear visual hierarchies. A region-specific color theme was adopted to make geographic identification intuitive: Central (Dark Blue), East (Light Blue), South (Gray), and West (Purple / Pink).

3.2 Dashboard Structure and Layout

Section	Visual Type	Purpose
Left	KPI Cards	Show Revenue, Profit, Quantity with YoY comparisons
Middle	Area Line Chart	Show monthly sales trend from Jan to Dec
Right	Horizontal Bar Chart	Show sales breakdown by customer segment

3.3 Visual Types and Justification

- **KPI Cards:** Provide instant visibility into high-level metrics (Total Revenue, Total Profit, Total Quantity) alongside YoY growth indicators.
- **Area Line Charts:** Display monthly trends across the year to identify seasonal surges, particularly in Q4.
- **Horizontal Bar Charts:** Used for clear, ranked comparison of customer segment contributions.

3.4 Interactivity and Filters

Two primary slicers were added at the top of the dashboard for dynamic filtering: a **Metric Slicer** (Total Revenue / Total Profit / Total Quantity) to change the focus of the visuals, and a **Year Slicer** (2020 to 2023) to review specific annual performance.

3.5 Dashboard Screenshots

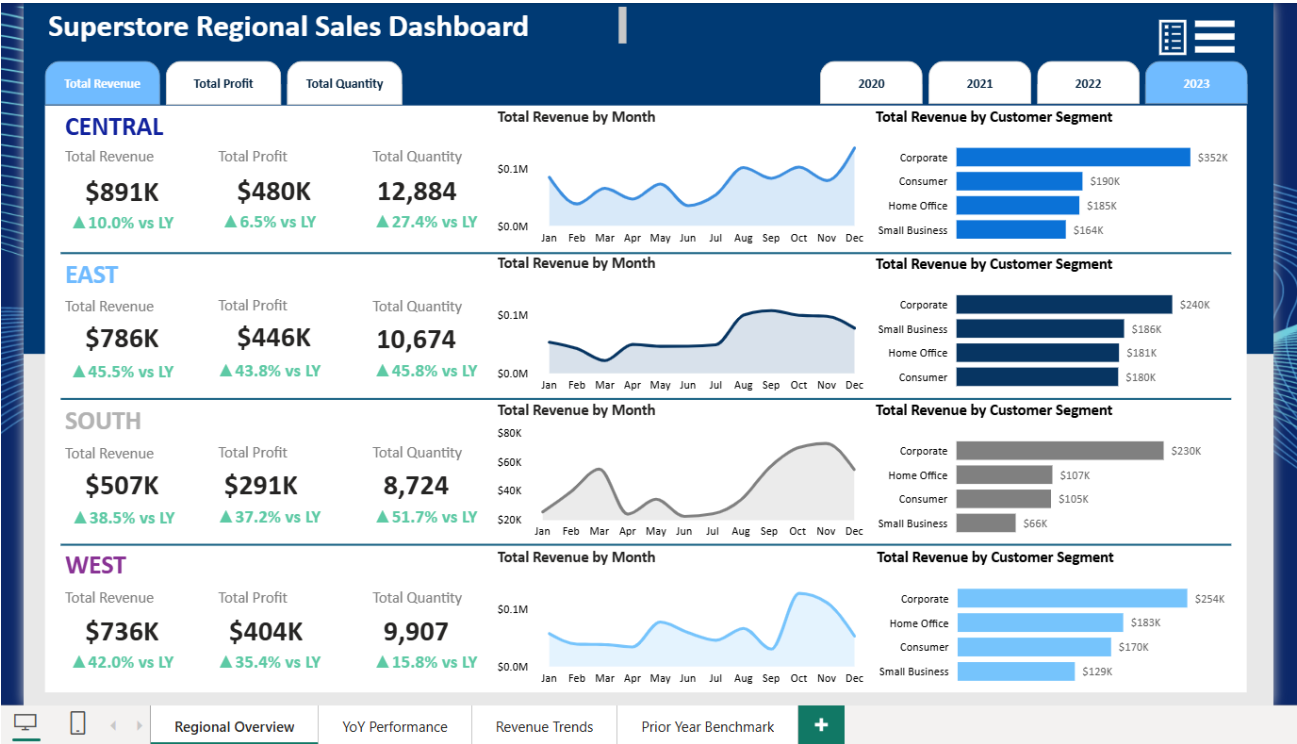


Figure 1: Regional Sales Overview

Figure 1: Regional Sales Overview: This dynamic primary dashboard features KPI cards, monthly trend area charts, and customer segment bar charts for each region. Users can interact with the data using the top slicers to filter by year and toggle between Revenue, Profit, and Quantity metrics.

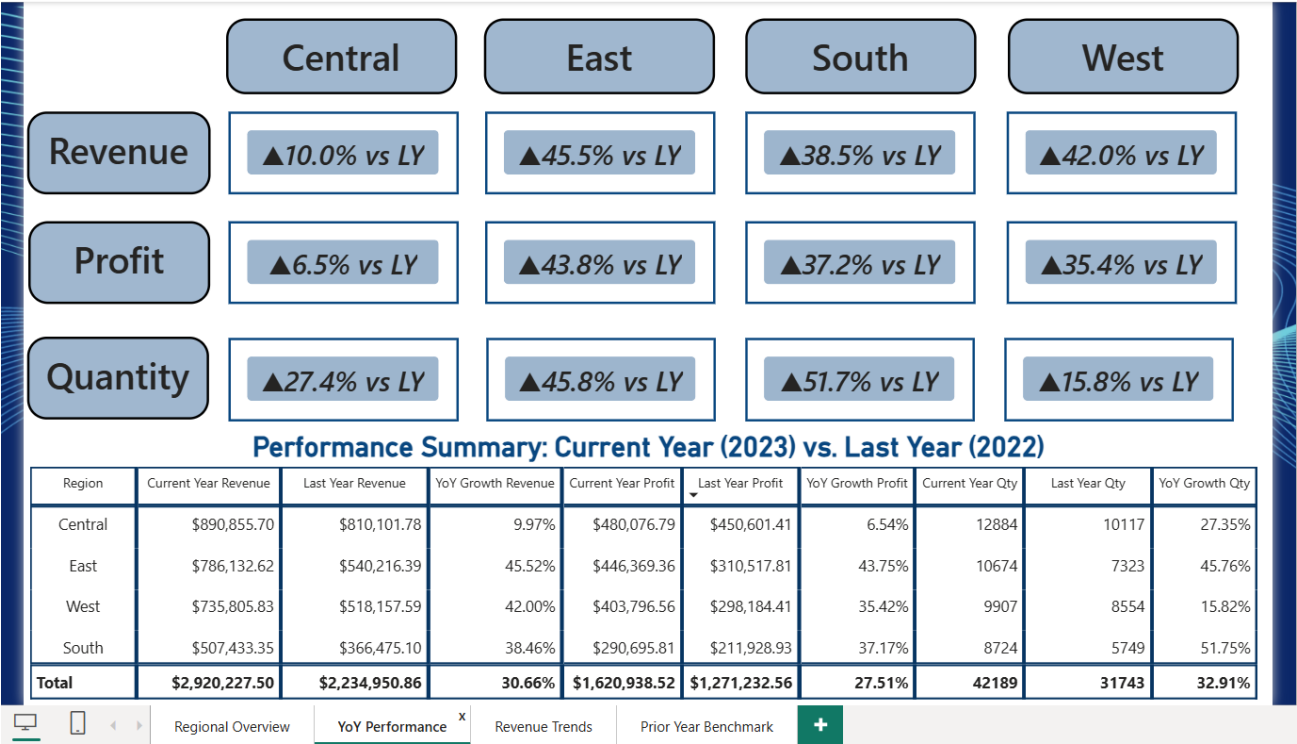


Figure 2: YoY Performance Matrix and Summary Table

Figure 2: YoY Performance Matrix and Summary Table: This static view provides a detailed breakdown of Year-over-Year growth percentages for Revenue, Profit, and Quantity across all regions, supported by a comprehensive summary table comparing 2022 and 2023 figures.

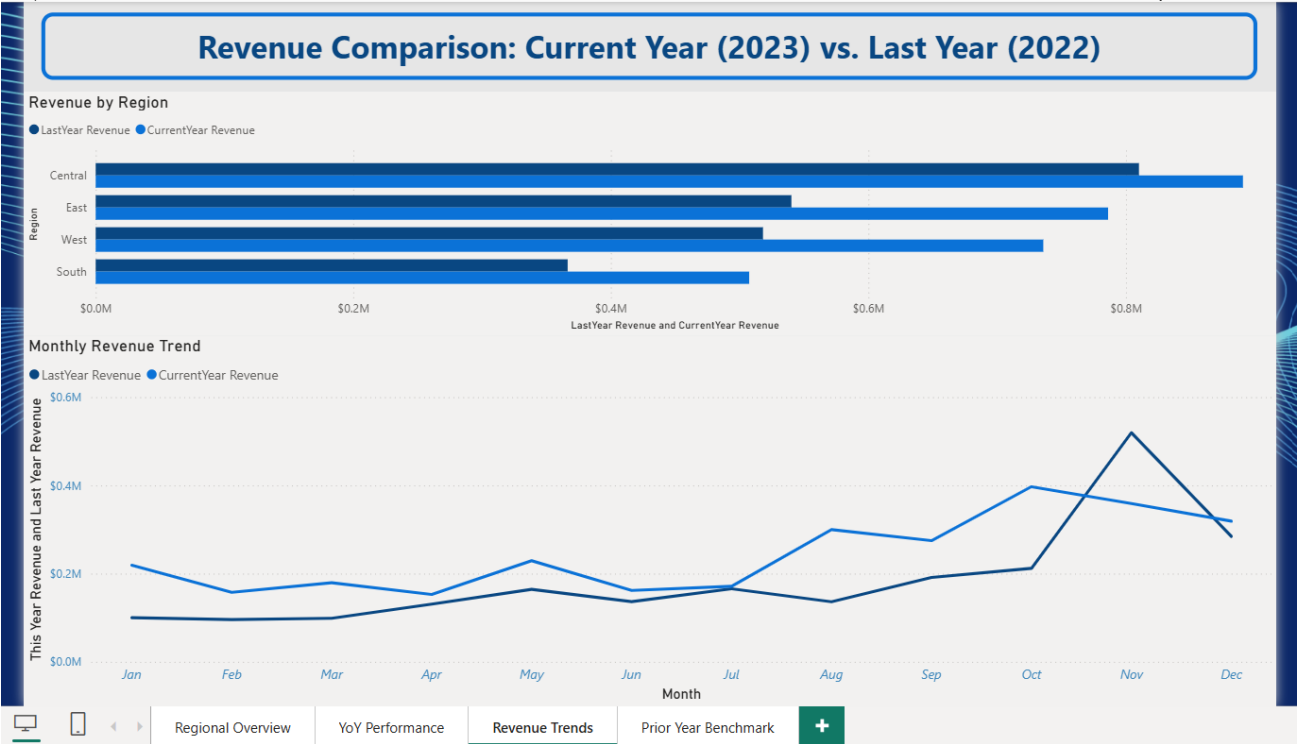


Figure 3: Revenue Comparison and Monthly Trends (CY vs. LY)

Figure 3: Revenue Comparison and Monthly Trends (CY vs. LY): Focused specifically on revenue, this static page compares the Current Year (2023) to the Last Year (2022) using a comparative bar chart for regional totals and a dual-line chart to visualize monthly seasonality.

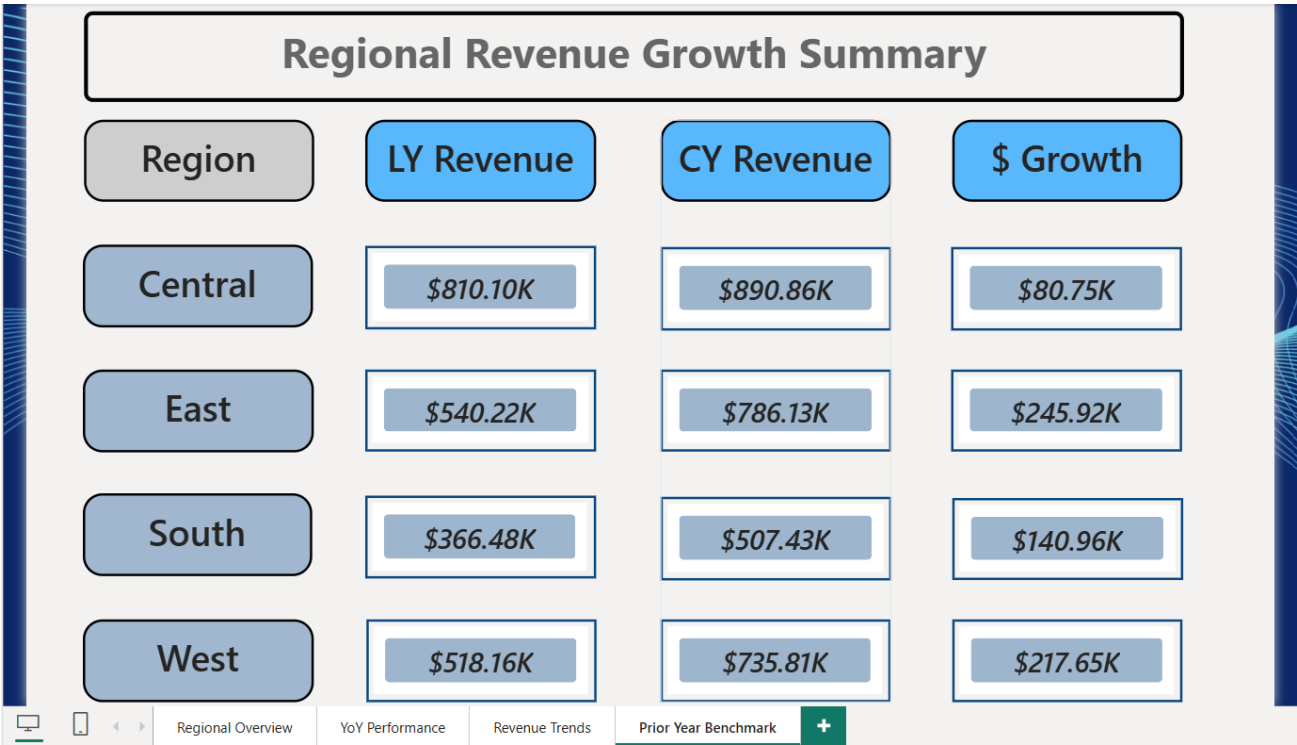


Figure 4: Regional Revenue Growth Summary

Figure 4: Regional Revenue Growth Summary: A focused summary table highlighting the absolute dollar growth (\$ Growth) between LY Revenue and CY Revenue across the Central, East, South, and West regions.

4. DAX Measures and Calculations

Document all significant DAX measures and calculated columns used in the Power BI report.

4.1 Measure Documentation

Measure Name	DAX Formula	Description
Total Revenue / Profit / Quantity	Total Revenue = SUM('Order Table'[Revenue])	Calculates the base sum of each primary metric.
LY Revenue / Profit / Quantity	LY Revenue = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Date Table'[Date]))	Calculates the previous year values using time intelligence.

Measure Name	DAX Formula	Description
YoY Growth %	Revenue YoY Growth = ([Total Revenue] / [LY Revenue]) - 1	Calculates the percentage growth compared to last year.
V2 Display Measures	VAR _p = [Revenue YoY Growth] VAR _a = IF(_p>=0, "▲", "▼") RETURN _a & FORMAT(ABS(_p), "0.0%") & " vs LY"	Formats YoY Growth into readable text with arrow indicators for KPI cards.
Conditional KPI Colors	Color1_Revenue = IF([Revenue YoY Growth] >= 0, "#5DCAA5", "#F09595")	Dynamic color formatting (Green for growth, Red for decline).
Static Current Year (MY_CY)	MY_CY Revenue = CALCULATE([Total Revenue], 'Order Table'[Year] = 2023)	Hardcodes the Current Year (2023) total to protect static visuals from page slicers.
Static Last Year (MY_LY)	MY_LY Revenue = CALCULATE([Total Revenue], 'Order Table'[Year] = 2022)	Hardcodes the Last Year (2022) total for direct historical comparison on static pages.
Static YoY % Growth	MY_YoY Revenue = DIVIDE([MY_CY Revenue] - [MY_LY Revenue], [MY_LY Revenue])	Calculates the exact percentage change between the static 2023 and 2022 values.

5. Key Findings and Insights

5.1 Finding 1: The Central Margin Squeeze

The Trend: The Central region brought in the highest overall Revenue (\$891K) and highest Quantity (12,884 items sold).

The Issue: While volume grew by **27.4%**, profit only ticked up by **6.5%**. This disproportionate lag indicates margin erosion—likely caused by heavy discounting, rising shipping costs, or a shift toward lower-margin products in this region.

5.2 Finding 2: West Region Shows Most Consistent Growth

The West region presents the cleanest trajectory in the dataset. It grew every single year across all three metrics (Revenue: \$372K to \$736K; Profit: \$206K to \$404K; Quantity: 5,477 to 9,907). By 2023, the region had nearly doubled its starting point. Like Central, it demonstrates strong seasonal spikes and a well-balanced mix of customer segments.

5.3 Finding 3: South Region Demonstrates Vulnerability

The South is the only region to experience multi-year profit declines (2020-2022) and flat quantity growth over the same period. While 2023 brought a sharp recovery (Revenue up to \$507K, Profit up to \$291K), the region remains the smallest overall. Furthermore, it is heavily dependent on Corporate clients, with Small Business contributing a fraction of the total, leaving the region exposed if corporate spending softens.

5.4 Finding 4: East Region Recovery and Balance

After a significant dip in 2021, the East region rebounded strongly to hit \$786K in revenue and \$446K in profit by 2023. Notably, in 2023, the East saw its Small Business profit reach \$110K—the highest of any region in any year. The tight spread between all four customer segments highlights a highly diversified and robust customer base.

5.5 Additional Observations

The Q4 Seasonal Surge: Regardless of the region's overall performance, the area line charts reveal a consistent and dramatic spike in revenue during November and December across all four years. This indicates that the Superstore's business model is highly seasonal and heavily dependent on year-end holiday purchasing cycles.

Dominance of the Corporate Segment: Across all regions, the Corporate customer segment consistently generates the highest share of both revenue and profit. While regions like the East and West have begun to balance this out with strong Consumer and Small Business sales, the Corporate segment remains the foundational pillar of the company's financial health.

6. Discussion

6.1 Interpretation of Results

The combination of Revenue, Profit, and Quantity metrics reveals that sustainable growth requires a balanced customer base. West led the charts because it captured high volumes without sacrificing profit margins, and they diversified their sales across Corporate, Consumer, Home Office, and Small Business segments. The South's struggle highlights the danger of segment over-reliance.

6.2 Comparison with Existing Knowledge

The analysis is limited by the lack of prior-year data before 2020, making initial YoY growth metrics appear artificially skewed. Geographic data stops at the regional level, masking state or city-level variations. Furthermore, without detailed cost structure data (overhead, logistics), it is impossible to determine if profit variations stem from pricing power or operational efficiency.

6.3 Limitations of the Analysis

Time Period Constraint: The dataset only covers the years 2020 through 2023. As a result, the analysis lacks a pre-pandemic baseline, making it difficult to compare these figures against long-term, normalized historical trends.

Lack of Cost Granularity: While overall profit and product margins are provided, the dataset is missing detailed operational and overhead cost breakdowns. This makes it impossible to determine whether regional profit squeezes are due to rising shipping costs, heavy discounting, or operational inefficiencies.

Geographic Depth: The geographic data provided stops at the broad regional level (Central, East, South, West). Because state or city-level data is unavailable, drill-downs are not possible, which masks localized performance trends and consumer behaviors.

7. Conclusion and Recommendations

7.1 Conclusion

As a whole, the Superstore showed growth across all regions from 2020 to 2023. Central grew into the top position through consistent performance and strong seasonal execution. The East rebounded with an impressively balanced customer base, while the West acted as the quiet achiever, growing steadily without a single down year. The South improved in 2023, but its over-reliance on Corporate clients remains a structural weakness.

7.2 Recommendations

- Apply the robust Q4 seasonal inventory strategies used by Central and West to the East and South to maximize year-end sales.
- Develop targeted marketing initiatives in the South region to attract Consumer and Small Business segments, reducing reliance on Corporate revenue.
- Investigate the root cause of the East's 2021 drop and its subsequent 2023 Small Business boom to replicate that mid-market success elsewhere.

7.3 Future Work

Future iterations of this dashboard should incorporate deeper geographic drill-downs (state/city level) to pinpoint localized performance. Adding product category context and tracking customer retention rates would also significantly enhance the strategic value of the reporting. Finally, integrating a simple trend-based forecasting model would enable proactive target setting.

8. Significant Contributions

8.1 Social Impact of the Project

The Superstore Regional Sales Performance Dashboard contributes to evidence-based retail management by making complex sales data accessible and interpretable to non-technical stakeholders. By enabling regional managers to quickly identify underserved markets and optimize resource allocation, this project promotes data literacy and empowers local teams to make decisions that improve organizational efficiency.

8.2 Linkage to UN Sustainable Development Goals (SDGs)

This project aligns with SDG 8: Decent Work and Economic Growth and SDG 9: Industry, Innovation & Infrastructure.

SDG Goal	Relevant Target(s)	How This Project Contributes
Primary SDG: SDG 8 — Decent Work & Economic Growth	Target 8.2	By creating an interactive BI dashboard, this project upgrades the technological capabilities of the retail business. It allows managers to identify unprofitable regions and scale successful strategies, directly leading to better economic productivity and sustainable business growth.
Secondary SDG: SDG 9 — Industry, Innovation & Infrastructure	Target 9.b	The use of Power BI demonstrates the application of digital innovation to improve business intelligence infrastructure and analytical capabilities.

8.3 Primary SDG Alignment

Select the most relevant SDG(s) and explain in detail how your project contributes to achieving them.

SDG Goal	Relevant Target(s)	How This Project Contributes
Primary SDG: SDG 8 — Decent Work & Economic Growth	Target 8.2: Achieve higher levels of economic productivity through diversification and technological upgrading.	By transitioning the company from static reporting to an interactive BI dashboard, this project significantly upgrades the technological capabilities of the retail business. It allows managers to identify unprofitable strategies and scale successful ones (like the East's Small Business growth), driving sustainable economic productivity.

SDG Goal	Relevant Target(s)	How This Project Contributes
Secondary SDG: SDG 12 — Responsible Consumption & Production	Target 12.5: Substantially reduce waste generation through prevention, reduction, recycling and reuse.	<i>The dashboard clearly visualizes the massive Q4 seasonal demand surges. By utilizing these trend visualizations, management can implement precise "just-in-time" inventory purchasing, directly reducing the likelihood of unsold overstock and physical retail waste.</i>
Optional SDG: SDG 9 — Industry, Innovation & Infrastructure	Target 9.b: Support domestic technology development, research and innovation.	<i>The implementation of Microsoft Power BI demonstrates the practical application of digital innovation to modernize legacy business intelligence infrastructure, fostering a culture of data literacy.</i>

8.4 Narrative Explanation

Our analysis of the Superstore regional sales dataset directly supports **SDG 8 (Decent Work and Economic Growth)** by revealing critical vulnerabilities and opportunities within the business, such as the South's over-reliance on Corporate clients and the East's booming Small Business sector. The interactive dashboard empowers regional directors to dynamically adjust their sales strategies, shift marketing focus to underserved segments, and prepare for end-of-year seasonal surges. By making this complex financial information accessible and visually intuitive, we promote enterprise-wide data literacy and evidence-based decision-making. These are the foundational requirements for building resilient businesses that contribute to sustainable economic growth in their local communities.

9. Group Member Contributions

Name	Role(s)	Tasks Completed
NHEN Theary	Project Lead / Writer	<i>Coordinated the overall project, wrote the Introduction and Conclusion sections, and compiled the final report.</i>
PHYRUN Pichchorda	<i>Data Engineer</i>	<i>Handled data preprocessing in Power Query, removed duplicates, fixed data types, and created the Date Table</i>
SAMBATH Seakty	<i>Dashboard Designer</i>	<i>Designed the main dashboard layout, set up the regional color themes, and built the bar and line charts.</i>
NOV Panhavath	<i>Data Analyst / Writer</i>	<i>Wrote the DAX measures (Total Revenue, YoY Growth logic), analyzed the regional data, and drafted the key findings.</i>
SOPHAT Vitou	<i>Dashboard Designer</i>	<i>Created the interactive KPI cards, set up the Year and Metric slicers, and ensured cross-filtering worked.</i>
NOB Sreynich	<i>Report Writer / Reviewer</i>	<i>Wrote the Social Impact and SDG sections, reviewed the data for accuracy, and formatted the final document.</i>

References

- [1] Superstore Regional Sales Dashboard (2020–2023). Internal business intelligence report. Dashboard data provided by the course instructor. Available: <https://www.kaggle.com/>
- [2] Microsoft, "Power BI documentation," Microsoft Learn. Accessed: 2026. [Online]. Available: <https://learn.microsoft.com/en-us/power-bi/>
- [3] S. Few, Show Me the Numbers: Designing Tables and Graphs to Enlighten, 2nd ed. Analytics Press, 2012.
- [4] T. H. Davenport, and J. G. Harris, Competing on Analytics: The New Science of Winning. Harvard Business School Press, 2007.
- [5] United Nations, "The 17 Goals," Sustainable Development Goals. [Online]. Available: <https://sdgs.un.org/goals>

Appendix

Appendix A: Full DAX Measure Code

```
// Fact Table Calculated Columns
Revenue = 'Order Table'[Unit Price] * 'Order Table'[Quantity]
Profit = 'Order Table'[Revenue] * 'Order Table'[Product Margin]
Month_Number = MONTH('Order Table'[Order Date])
Month_Name = FORMAT('Order Table'[Order Date], "mmm")
Year = YEAR('Order Table'[Order Date])

// Base Dynamic Measures
Total Revenue = SUM('Order Table'[Revenue])
Total Profit = SUM('Order Table'[Profit])
Total Quantity = SUM('Order Table'[Quantity])

// Time Intelligence Dynamic Measures
LY Revenue = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Date Table'[Date]))
LY Profit = CALCULATE([Total Profit], SAMEPERIODLASTYEAR('Date Table'[Date]))
LY Quantity = CALCULATE([Total Quantity], SAMEPERIODLASTYEAR('Date Table'[Date]))

Revenue YoY Growth = ([Total Revenue] / [LY Revenue]) - 1
Profit YoY Growth = ([Total Profit] / [LY Profit]) - 1
Quantity YoY Growth = ([Total Quantity] / [LY Quantity]) - 1

// Conditional Formatting Measures (Colors)
Color1_Revenue = IF([Revenue YoY Growth] >= 0, "#5DCAA5", "#F09595")
Color2_Profit = IF([Profit YoY Growth] >= 0, "#5DCAA5", "#F09595")
Color3_Quantity = IF([Quantity YoY Growth] >= 0, "#5DCAA5", "#F09595")

// Visual Indicator Measures (Arrows & Text)
V2_Revenue YoY Growth =
    VAR _p = [Revenue YoY Growth]
    VAR _a = IF(_p>=0, "▲", "▼")
    RETURN _a & FORMAT(ABS(_p), "0.0%") & " vs LY"

V2_Profit YoY Growth =
    VAR _p = [Profit YoY Growth]
    VAR _a = IF(_p>=0, "▲", "▼")
    RETURN _a & FORMAT(ABS(_p), "0.0%") & " vs LY"

V2_Quantity YoY Growth =
    VAR _p = [Quantity YoY Growth]
    VAR _a = IF(_p>=0, "▲", "▼")
    RETURN _a & FORMAT(ABS(_p), "0.0%") & " vs LY"

// Static Measures (Current Year = 2023)
MY_CY Profit = CALCULATE([Total Profit], 'Order Table'[Year] = 2023)
MY_CY Quantity = CALCULATE([Total Quantity], 'Order Table'[Year] = 2023)
```

MY_CY Revenue = CALCULATE([Total Revenue], 'Order Table'[Year] = 2023)

// Static Measures (Last Year = 2022)

MY_LY Profit = CALCULATE([Total Profit], 'Order Table'[Year] = 2022)

MY_LY Quantity = CALCULATE([Total Quantity], 'Order Table'[Year] = 2022)

MY_LY Revenue = CALCULATE([Total Revenue], 'Order Table'[Year] = 2022)

// Static YoY Growth Measures

MY_YoY Profit = DIVIDE([MY_CY Profit] - [MY_LY Profit], [MY_LY Profit])

MY_YoY Quantity = DIVIDE([MY_CY Quantity] - [MY_LY Quantity], [MY_LY Quantity])

MY_YoY Revenue = DIVIDE([MY_CY Revenue] - [MY_LY Revenue], [MY_LY Revenue])

Appendix B: Additional Dashboard Screenshots

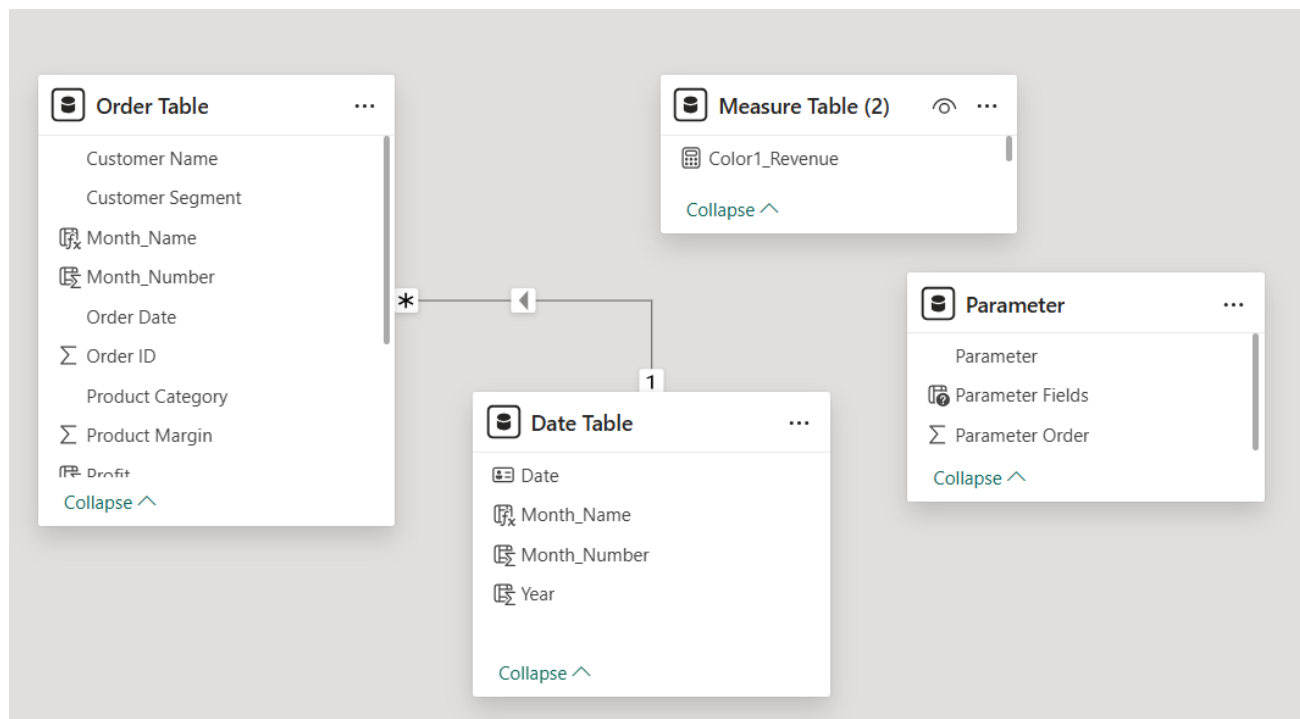


Figure 5: Power BI Data Model Architecture

Figure 5: Power BI Data Model Architecture: This view illustrates the single one-to-many relationship between the Date Table and Order Table required for time intelligence. The Measure Table and Parameter table remain intentionally disconnected as independent supporting components for calculations and dynamic metric switching.

Appendix C: Raw Data Sample

Order ID	Product Category	Region	Quantity	Unit Price	Product Margin	Order Date
37537	Chairs & Chairmats	Central	43	100.98	69.0%	1/2/2020
37537	Storage & Organization	South	32	155.06	59.0%	1/2/2020
87946	Storage & Organization	East	8	155.06	59.0%	1/2/2020
37537	Chairs & Chairmats	South	4	291.73	56.0%	1/2/2020
90853	Paper	West	4	9.11	40.0%	1/2/2020
90853	Paper	South	11	15.04	39.0%	1/2/2020
9285	Binder Accessories	South	3	40.98	36.0%	1/2/2020
89083	Binder Accessories	South	1	40.98	36.0%	1/2/2020
89999	Office Furnishings	South	3	20.98	78.0%	1/3/2020
87463	Computer Peripherals	Central	3	43.22	66.0%	1/3/2020
88296	Tables	West	1	212.60	64.0%	1/3/2020
90430	Office Machines	East	1	213.45	59.0%	1/3/2020
90337	Telephones	Central	2	110.99	57.0%	1/3/2020
7909	Pens & Art Supplies	Central	29	4.26	44.0%	1/3/2020
89583	Pens & Art Supplies	South	7	4.26	44.0%	1/3/2020
90337	Paper	South	4	5.28	40.0%	1/3/2020
87463	Office Machines	West	12	574.74	37.0%	1/3/2020
87463	Paper	East	3	10.14	36.0%	1/3/2020
86356	Chairs & Chairmats	West	1	350.98	61.0%	1/4/2020

Figure 6: Raw Dataset

Figure 6: Raw Dataset Sample: A snapshot of the initial Superstore transactional data prior to preprocessing. This view displays the foundational columns—such as Quantity, Unit Price, and Product Margin—which were subsequently used to generate the calculated Revenue and Profit measures within the Power BI data model.